实验名称：SQL 基础操作实验

作者：郭威威

学号：2414308

指导老师：李旭东

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目录

实验目的

掌握 SQL 语言的基本语法，熟悉数据库的创建、表结构设计、数据插入与查询操作，理解关系型数据库的逻辑结构。

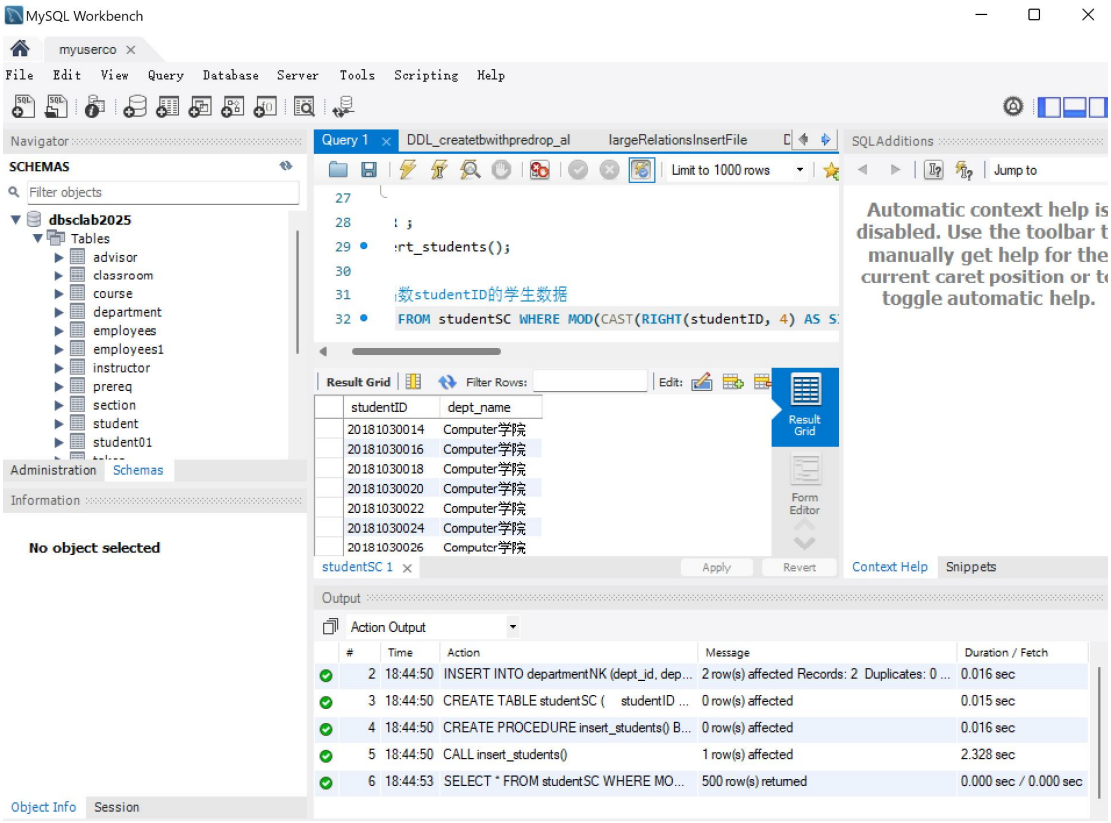
具体包括：

- 基本查询与自然连接
- 重命名操作与字符串处理
- 元组排序与集合操作
- 聚合函数与嵌套子查询

实验内容与步骤

QUIZ 1 ：

1. 创建 departmentNK 表，假设字段合理，如学院相关。
2. 插入学院数据。
3. 创建 studentSC 表，包含 studentID 等字段。
4. 插入 1000 条数据，ID 范围正确，学院为 Computer 学院。
5. 查询偶数 ID 的学生。



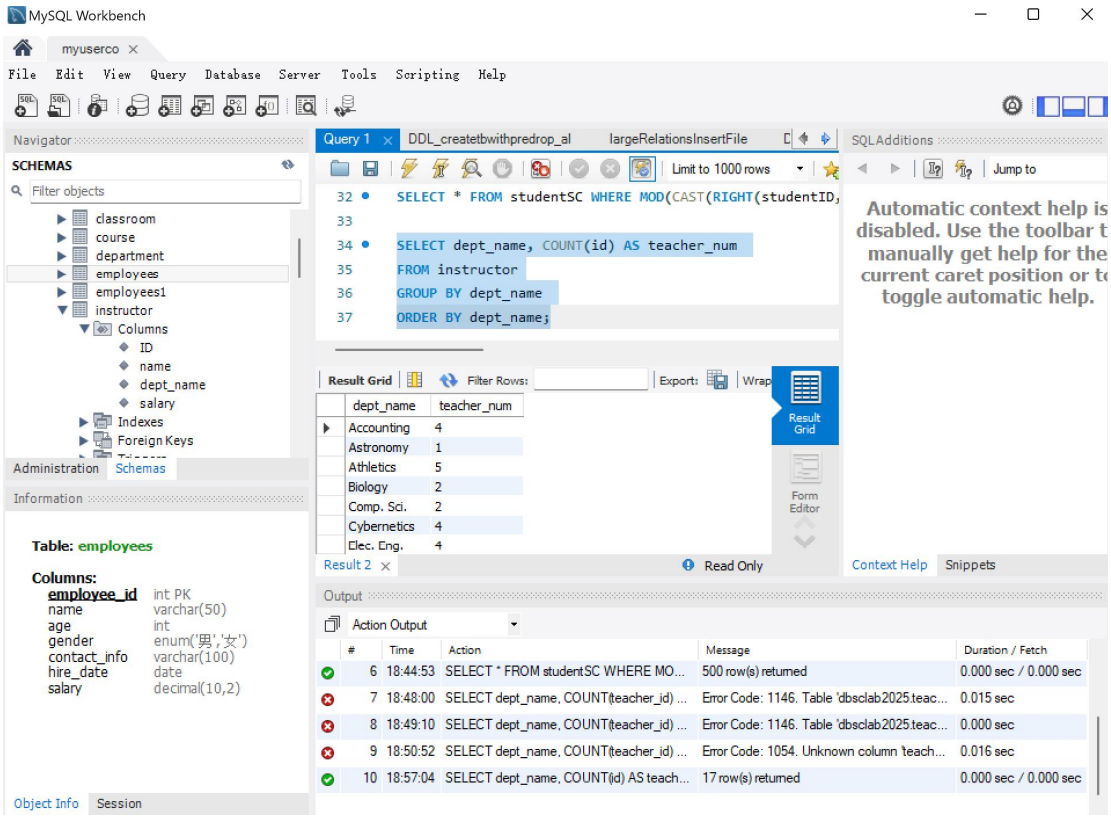
代码如下 1

```sql
-- 创建 departmentNK 表
CREATE TABLE departmentNK (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);

-- 插入学院数据
INSERT INTO departmentNK (dept_id, dept_name) VALUES
(1, 'Computer 学院'), (2, '其他学院');

-- 创建 studentSC 表
CREATE TABLE studentSC (
    studentID VARCHAR(20) PRIMARY KEY,
    dept_name VARCHAR(50)
);

-- 插入 1000 条学生数据
DELIMITER $$
CREATE PROCEDURE insert_students()
BEGIN
    DECLARE i INT DEFAULT 1;
```

```
    WHILE i <= 1000 DO
        INSERT INTO studentSC (studentID, dept_name)
        VALUES (CONCAT('2018103', LPAD(i, 4, '0')), 'Computer 学院
');
        SET i = i + 1;
    END WHILE;
END $$
DELIMITER ;
CALL insert_students();

-- 显示偶数 studentID 的学生数据
SELECT * FROM studentSC WHERE MOD(CAST(RIGHT(studentID, 4) AS
SIGNED), 2) = 0;
```

QUIZ 2：

1. 分组统计学院教师数，排序。

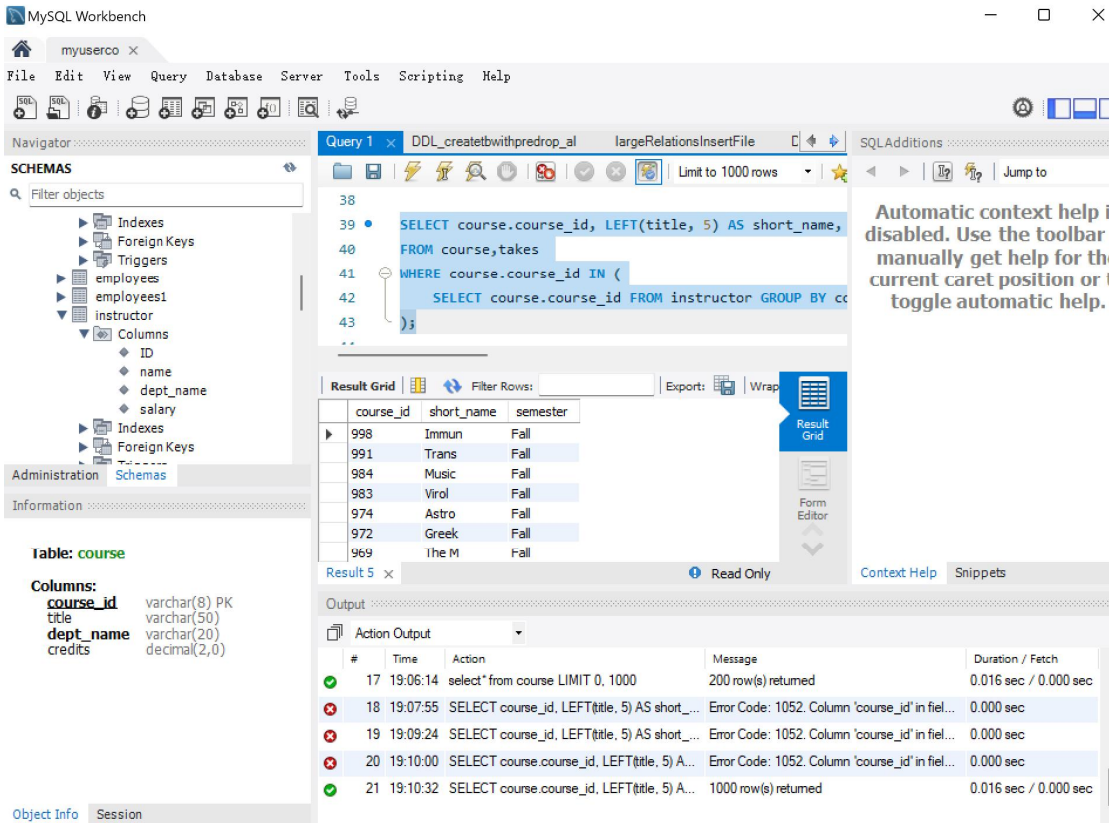


代码

```sql
SELECT dept_name, COUNT(id) AS teacher_num
FROM instructor
```

```
GROUP BY dept_name
ORDER BY dept_name;
```

2. 统计多教师授课的课程，取课程名前 5 字符。



代码

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-- 创建 departmentNK 表
CREATE TABLE departmentNK (
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-- 插入学院数据
INSERT INTO departmentNK (dept_id, dept_name) VALUES
(1, 'Computer 学院'), (2, '其他学院');

-- 创建 studentSC 表
CREATE TABLE studentSC (
    studentID VARCHAR(20) PRIMARY KEY,
    dept_name VARCHAR(50)
);
```

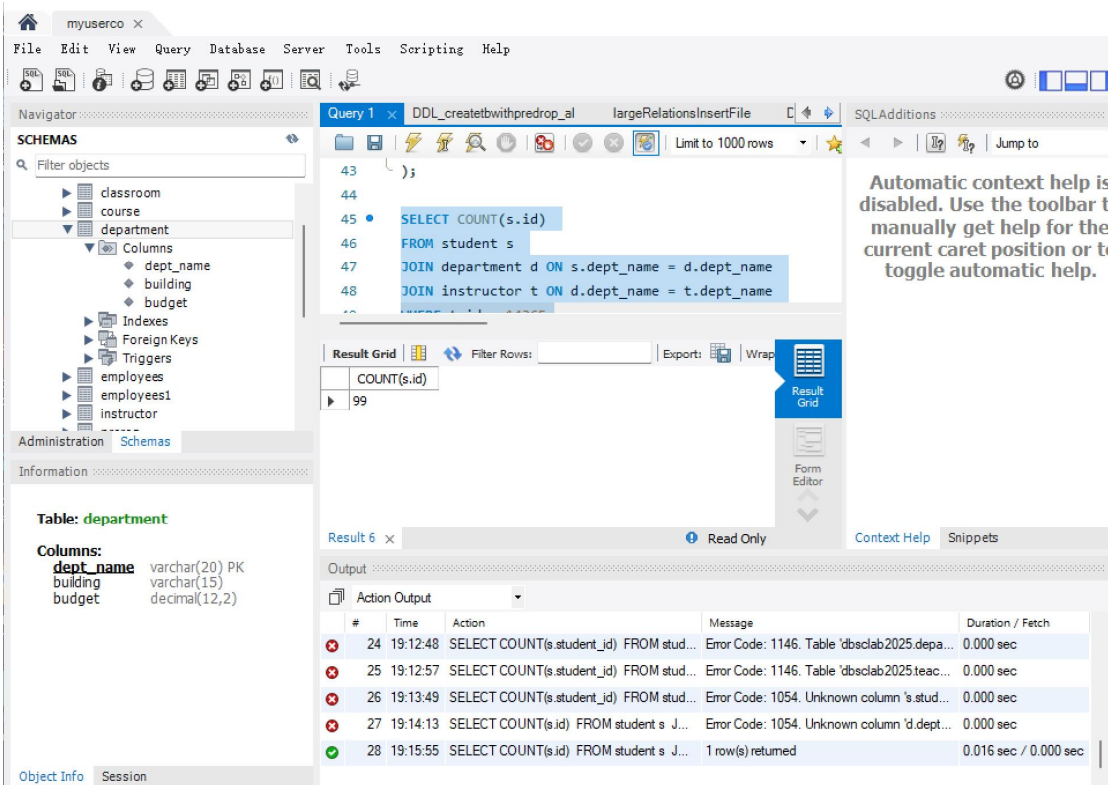
```
-- 插入 1000 条学生数据
DELIMITER $$
CREATE PROCEDURE insert_students()
BEGIN
    DECLARE i INT DEFAULT 1;
    WHILE i <= 1000 DO
        INSERT INTO studentSC (studentID, dept_name)
        VALUES (CONCAT('2018103', LPAD(i, 4, '0')), 'Computer 学院
');
        SET i = i + 1;
    END WHILE;
END $$
DELIMITER ;
CALL insert_students();

-- 显示偶数 studentID 的学生数据
SELECT * FROM studentSC WHERE MOD(CAST(RIGHT(studentID, 4) AS
SIGNED), 2) = 0;

SELECT dept_name, COUNT(id) AS teacher_num
FROM instructor
GROUP BY dept_name
ORDER BY dept_name;

SELECT course.course_id, LEFT(title, 5) AS short_name, semester
FROM course,takes
WHERE course.course_id IN (
    SELECT course.course_id FROM instructor GROUP BY
course.course_id HAVING COUNT(id) > 1
);
```

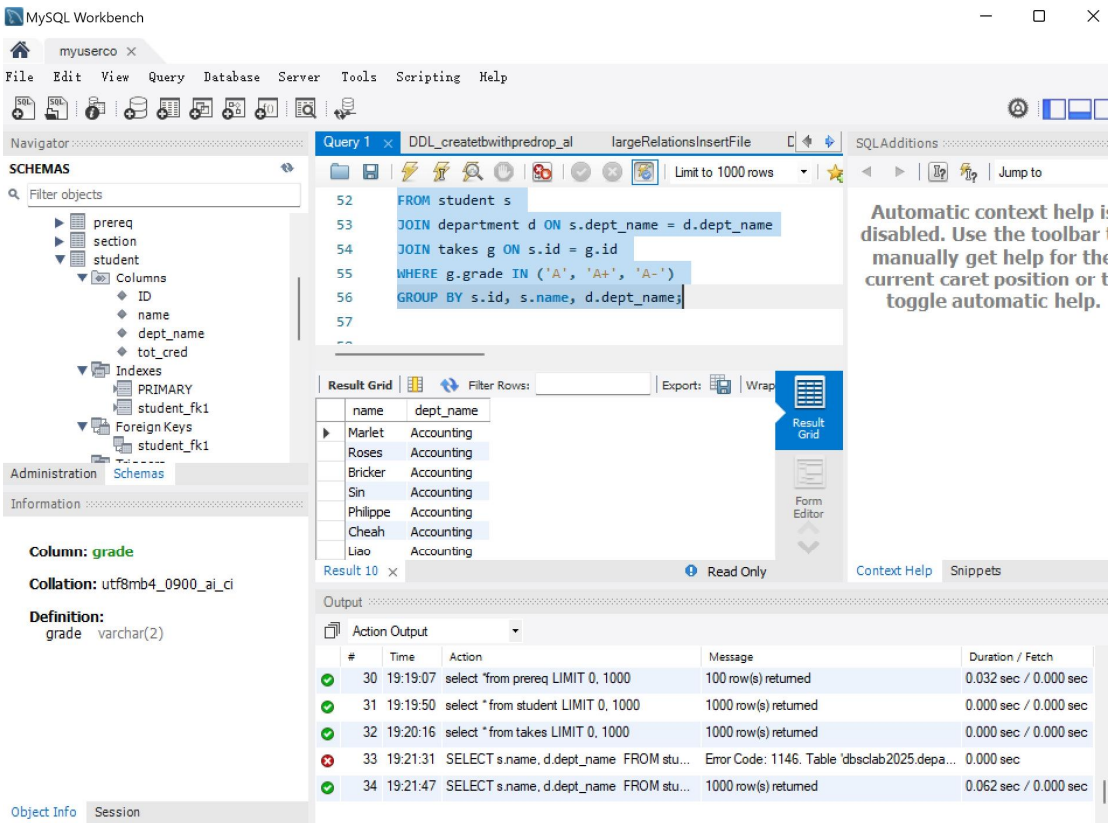
3. 通过教师 ID 找学院，统计该学院学生数。



代码

```sql
SELECT COUNT(s.id)
FROM student s
JOIN department d ON s.dept_name = d.dept_name
JOIN instructor t ON d.dept_name = t.dept_name
WHERE t.id = 14365;
```

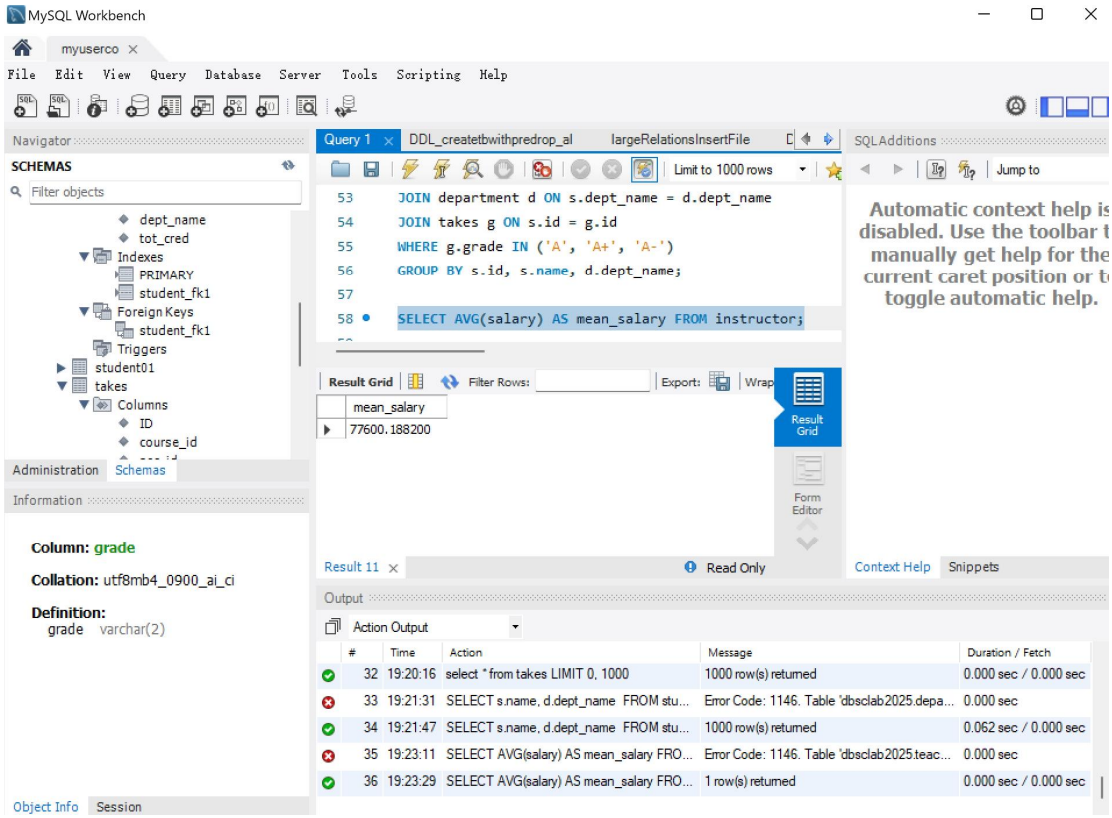
4. 筛选成绩等级，关联学生和学院。

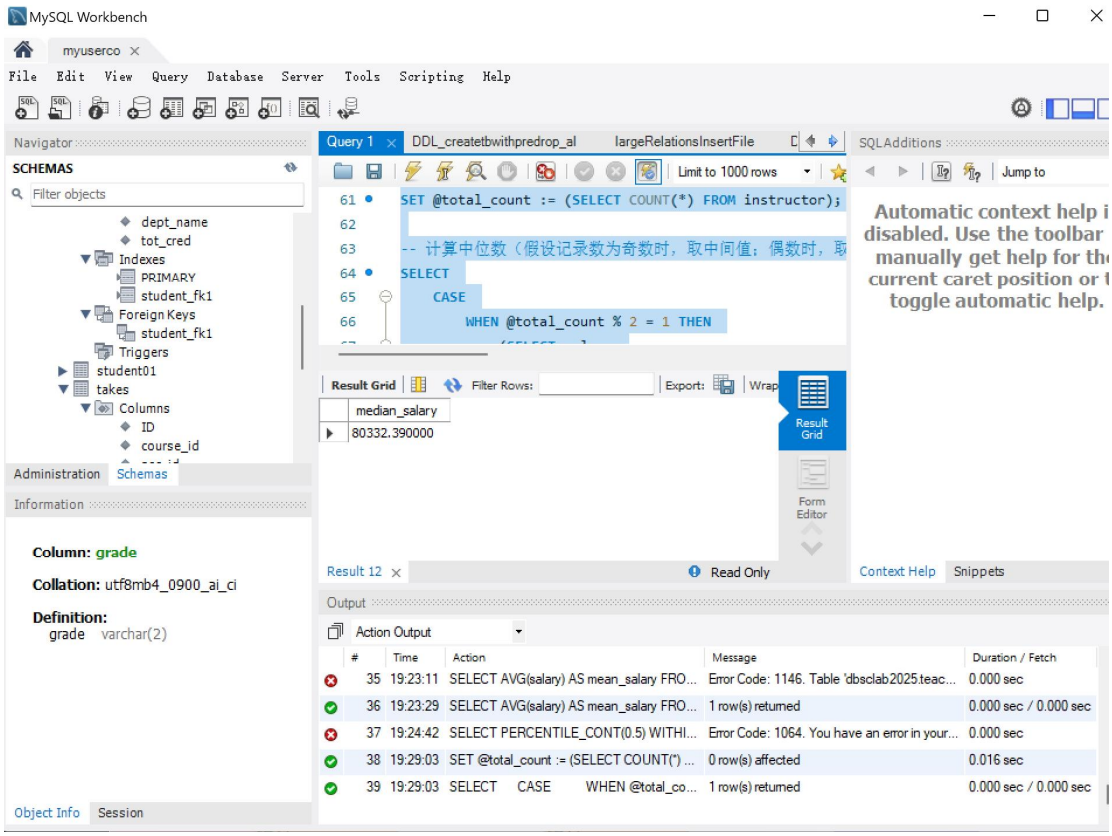


代码

```sql
SELECT s.name, d.dept_name
FROM student s
JOIN department d ON s.dept_name = d.dept_name
JOIN takes g ON s.id = g.id
WHERE g.grade IN ('A', 'A+', 'A-')
GROUP BY s.id, s.name, d.dept_name;
```

5. 计算工资均值和中位数，注意中位数在 SQL 中的实现。


MySQL Workbench
myuserco
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
dept_name
tot_cred
Indexes
PRIMARY
student_fk1
Foreign Keys
student_fk1
Triggers
student01
takes
Columns
ID
course_id
Administration Schemas
Information
Column: grade
Collation: utf8mb4_0900_ai_ci
Definition:
grade varchar(2)
Object Info Session
Query 1 DDL_createtbwithpredrop_al largeRelationsInsertFile
Limit to 1000 rows
53 JOIN department d ON s.dept_name = d.dept_name
54 JOIN takes g ON s.id = g.id
55 WHERE g.grade IN ('A', 'A+', 'A-')
56 GROUP BY s.id, s.name, d.dept_name;
57
58 SELECT AVG(salary) AS mean_salary FROM instructor;
Result Grid Filter Rows: Export: Wrap
mean_salary
77600.188200
Result 11
Read Only
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar t manually get help for the current caret position or t toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
32 19:20:16 select * from takes LIMIT 0, 1000 1000 row(s) returned 0.000 sec / 0.000 sec
33 19:21:31 SELECT s.name, d.dept_name FROM stu... Error Code: 1146. Table 'dbsclab2025.depa... 0.000 sec
34 19:21:47 SELECT s.name, d.dept_name FROM stu... 1000 row(s) returned 0.062 sec / 0.000 sec
35 19:23:11 SELECT AVG(salary) AS mean_salary FRO... Error Code: 1146. Table 'dbsclab2025.teac... 0.000 sec
36 19:23:29 SELECT AVG(salary) AS mean_salary FRO... 1 row(s) returned 0.000 sec / 0.000 sec


MySQL Workbench
myuserco
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
dept_name
tot_cred
Indexes
PRIMARY
student_fk1
Foreign Keys
student_fk1
Triggers
student01
takes
Columns
ID
course_id
Administration Schemas
Information
Column: grade
Collation: utf8mb4_0900_ai_ci
Definition:
grade varchar(2)
Object Info Session
Query 1 DDL_createtbwithpredrop_al largeRelationsInsertFile
Limit to 1000 rows
61 SET @total_count := (SELECT COUNT(*) FROM instructor);
62
63 -- 计算中位数（假设记录数为奇数时，取中间值；偶数时，取
64 SELECT
65 CASE
66 WHEN @total_count % 2 = 1 THEN
Result Grid Filter Rows: Export: Wrap
median_salary
80332.390000
Result 12
Read Only
SQLAdditions
Jump to
Automatic context help i disabled. Use the toolbar manually get help for th current caret position or t toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
35 19:23:11 SELECT AVG(salary) AS mean_salary FRO... Error Code: 1146. Table 'dbsclab2025.teac... 0.000 sec
36 19:23:29 SELECT AVG(salary) AS mean_salary FRO... 1 row(s) returned 0.000 sec / 0.000 sec
37 19:24:42 SELECT PERCENTILE_CONT(0.5) WITHI... Error Code: 1064. You have an error in your... 0.000 sec
38 19:29:03 SET @total_count := (SELECT COUNT(*) ... 0 row(s) affected 0.016 sec
39 19:29:03 SELECT CASE WHEN @total_co... 1 row(s) returned 0.000 sec / 0.000 sec

代码

```sql
```

```
#计算平均值
SELECT AVG(salary) AS mean_salary FROM instructor;

#计算中位数
-- 计算总记录数
SET @total_count := (SELECT COUNT(*) FROM instructor);

-- 计算中位数（假设记录数为奇数时，取中间值；偶数时，取中间两个值的平均值）
SELECT
    CASE
        WHEN @total_count % 2 = 1 THEN
            (SELECT salary
             FROM (SELECT salary, @rownum := @rownum + 1 AS rownum
                   FROM instructor, (SELECT @rownum := 0) r
                   ORDER BY salary) sub
             WHERE sub.rownum = (@total_count + 1) / 2)
        ELSE
            (SELECT AVG(salary)
             FROM (SELECT salary, @rownum := @rownum + 1 AS rownum
                   FROM instructor, (SELECT @rownum := 0) r
                   ORDER BY salary) sub
             WHERE sub.rownum IN (@total_count / 2, (@total_count / 2) + 1))
    END AS median_salary;
```

6. 分析两个 SQL 语句的区别，主要看 JOIN 条件位置。

语句不同。a) 中 ON 条件是 student.ID = takes.ID，正确关联；b) 中 ON true，然后 WHERE student.ID = takes.ID，会导致 LEFT JOIN 变成 INNER JOIN 效果，因为 WHERE 过滤了 null，丢失左表不匹配的数据。

a 语句：ON student.ID = takes.ID 正确关联两表，保留左表（student）所有记录。

b 语句：ON true 使连接无过滤，WHERE student.ID = takes.ID 会过滤掉左表不匹配的 null，实际效果等同于内连接（丢失左表未匹配数据）。

实验结论

通过本次实验，熟练掌握了 SQL 的核心操作，包括复杂查询、多表连接及聚合函数的应用。实验结果验证了关系型数据库的逻辑设计原则，为后续高级 SQL 技术（如事务处理、索引优化）奠定了基础。